SelectBoost, a General Algorithm to Enhance the Performance of Variable Selection in Biological Network Reverse Engineering

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Conclusion

We created SelectBoost (Bertrand, 2020) a new general algorithm which improves the precision of any existing variable selection method.
This algorithm is based
- on highly intensive simulations and
- takes into account the correlation structure of the data.

Our algorithm can either produce
- a confidence index for variable selection or
- be used to design experiments.

We showed the performance of our algorithm on both simulated and real data and apply it in two different ways to improve biological network reverse-engineering, see **Fig 1** for one use case.

Availability

`https://fbertran.github.io/SelectBoost/`

or as the `selectboost` CRAN package

Functions

1. **simulation_DATA** to simulate data for benchmarking
2. **fasboost** performs by default 100 model resamplings the using the variable selection function to be used to assess variable memberships. Defaults to lasso_msgps_AICc but many others lasso, elastinet, logistic glmnet are provided
3. **force.non.inc** returns a vector after ensuring that the proportion of times each variable was selected is non increasing with respect to the 1-c0 value

```
result.boost = force.non.inc(result.boost.raw)
result.boost
#>                 1    2    3 4 5 6 7 8    9 10
#>              1.00 1.00 1.00 0 0 0 0 0 1.00  0
#> c0 = 0.872 1.00 1.00 1.00 0 0 0 0 0 1.00  0
#> c0 = 0.827 1.00 1.00 1.00 0 0 0 0 0 1.00  0
#> c0 = 0.814 0.99 1.00 1.00 0 0 0 0 0 1.00  0
#> c0 = 0.451 0.96 1.00 0.96 0 0 0 0 0 0.96  0
#> c0 = 0.382 0.96 0.77 0.96 0 0 0 0 0 0.96  0
#> c0 = 0.14  0.96 0.77 0.95 0 0 0 0 0 0.66  0
#> c0 = 0.091 0.83 0.73 0.93 0 0 0 0 0 0.66  0
#> c0 = 0.065 0.83 0.73 0.84 0 0 0 0 0 0.66  0
#> c0 = 0.05  0.83 0.50 0.84 0 0 0 0 0 0.66  0
#> c0 = 0.03  0.83 0.50 0.82 0 0 0 0 0 0.66  0
#> c0 = 0.006 0.83 0.50 0.82 0 0 0 0 0 0.66  0
#> c0 = 0     0.68 0.50 0.66 0 0 0 0 0 0.57  0
```

Setting and Aim

Variable selection is a commonly required step

Technological innovations make it possible to measure large amounts of data in a single observation. As a consequence, problems in which the number P of variables is larger than the number N of observations have become common. As reviewed by Fan and Li (Fan and Li, 2006), such situations arise in many fields from fundamental sciences to social science, and variable selection is required to tackle these issues.

Penalized likelihood

There is a vast literature dealing with the problem of variable selection in both statistical and machine learning areas (Fan and Lv, 2010). The main variable selection methods can be gathered in the common framework of penalized likelihood. The estimate $\hat{\boldsymbol{\beta}}$ is then given by:

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}\in\mathbb{R}^P}{\arg\min}\left[-\ell_N(\boldsymbol{\beta}) + \sum_{p=1}^{P}\mathrm{pen}_\lambda(\beta_p)\right]$$

where $\ell_N(.)$ is the log-likelihood function, $\mathrm{pen}_\lambda(.)$ is a penalty function and $\lambda \in \mathbb{R}$ is the regularization parameter. As the goal is to obtain a sparse estimation of the vector of parameters β, a natural choice for the penalty function is to use the so-called $\mathcal{L}_0$ norm ($\|.\|_0$) which corresponds to the number of non-vanishing elements of a vector:

$$\begin{array}{rcl} \mathrm{pen}_\lambda : \mathbb{R} & \mapsto & \{0,\lambda\} \\ x & \mapsto & \begin{cases} \mathrm{pen}_\lambda(x) = \lambda & \text{if } x \neq 0 \\ \mathrm{pen}_\lambda(x) = 0 & \text{else} \end{cases} \end{array}$$

which induces $\sum_{p=1}^{P}\mathrm{pen}_\lambda(\beta_p) = \lambda\|\boldsymbol{\beta}\|_0$.
For example, when $\lambda = 1$, we get the Akaike Information Criterion (AIC, Akaike1974) and when $\lambda = \frac{\log(N)}{2}$ we get the Bayesian Information Criterion (BIC, Schwartz 1978).

Failure due to linear correlation

The goal of identifying the correct support of the regression is complicated and the reason why variable selection methods fail to select the set of non-zero variables can be summarized in two words: linear correlation.

Notations

N the number of observations	P the number of variables
$\mathbf{y} = (y_1, ..., y_N)$ the response variable	
$\mathbf{X} = (\mathbf{x}_{1.}, ..., \mathbf{x}_{P.})$ a variable matrix of size $N \times P$	
$\boldsymbol{\beta} = (\beta_1, ..., \beta_P)'$ a vector of parameters	

Method

Overcome linear correlation

The SelectBoost algorithm has been designed in a general framework to avoid to select non-predictive correlated features. The main goal is to improve the PPV, i.e. the proportion of selected variables which truly belong to the model.

Correlated resampling

The main idea of our algorithm is to consider that any observed value of a group of linearly correlated variables of the $\mathbf{X}$ matrix is the independent realization of a given random function, which is then used to perturb the observed values of the relevant correlated variables.

Sherical distribution

As we assume that the variables are centered and that $\|\mathbf{x}_{p.}\|^2 = 1$ for $p = 1, ..., P$, we know that $\mathbf{x}_{p.} \in \mathcal{S}^{N-2}$.
As a consequence, we use a spherical distribution for resampling. For instance, a multivariate Gaussian distribution assumption for $\mathbf{X} = (\mathbf{x}_{1.}, ..., \mathbf{x}_{P.})$ advocates for the use a von Mises-Fisher distribution (Sra, 2012).

The SelectBoost algorithm

1. To use the SB algorithm, we need a grouping method gr_{c_0} depending on a user-provided constant $0 \leq c_0 \leq 1$. It must satisfy:
   $\forall p \in \{1, ...P\} : gr_1(p) = \{p\} \quad \text{and} \quad gr_0(p) = \{1, ...P\}$.
   Let $gr_{c_0}(p)$ be the set of all variables which are considered to be linked to the variable $\mathbf{x}_p$. $\mathbf{X}_{gr_{c_0}(p)}$ is the submatrix of $\mathbf{X}$ with the columns which indices are in $gr_{c_0}(p)$.
2. Furthermore, we need to have a selection method:
   $$select \quad : \quad \mathbb{R}^{N\times P} \times \mathbb{R}^N \quad \rightarrow \quad \{0,1\}^P$$
   which maps the design matrix $\mathbf{X}$ and the response variable $\mathbf{y}$ to a 0-1 vector of length P with 1 at position p if the method selects the variable p and 0 otherwise.
3. We then use the von Mises-Fisher distribution to generate replacement of the original variables by some simulations to create B new design matrices $\mathbf{X}^{(1)}, ..., \mathbf{X}^{(B)}$.
4. The SB algorithm then applies the variable selection method $select$ to each of these matrices and returns a vector of length P with the frequency of apparition of each variable. The frequency of apparition of variable $\mathbf{x}_{p.}$, noted ζ_p is assumed to be an estimator of the probability of $\mathbf{x}_{p.}$ being in the true model.
5. The choice of c_0 is crucial. The model might not perturbed enough (too large c_0) or the variables are chosen at random (too small c_0).

The SB algorithm (**Algorithm 1**) returns the vector $\boldsymbol{\zeta} = (\zeta_1, ..., \zeta_P)'$. Compare these values $\boldsymbol{\zeta} = (\zeta_1, ..., \zeta_P)'$ to a threshold $\zeta_{\min}$ to determine which variables are selected.

Pseudo-code

Algorithm 1 Pseudo-code for the SelectBoost algorithm

Require: $gr_{c_0}, select, B, c_0$
  $\boldsymbol{\zeta} \leftarrow \mathbf{0_P}$
  for $b = 1, ..., B$ **do**
    $\mathbf{X}^{(b)} \leftarrow \mathbf{X}$
    for $p = 1, ..., P$ **do**
      $\mathbf{x}^{(b)}_{\mathbf{p}.} \leftarrow \phi^{-1}\left(\text{random-vMF}\left(\hat{\boldsymbol{\mu}}(\phi(\mathbf{X}_{\mathbf{gr_{c_0}(p)}})), \hat{\kappa}(\phi(\mathbf{X}_{\mathbf{gr_{c_0}(p)}}))\right)\right)$
    end for
    $\boldsymbol{\zeta} \leftarrow \boldsymbol{\zeta} + select(\mathbf{X}^{(b)}, \mathbf{y})$
  end for
  $\boldsymbol{\zeta} \leftarrow \boldsymbol{\zeta}/B$

Simulation studies

We combined the SB algorithm with our patterns CRAN R package designed to reverse engineer biological networks. This improved F-scores on simulated data, see **Fig 1**.

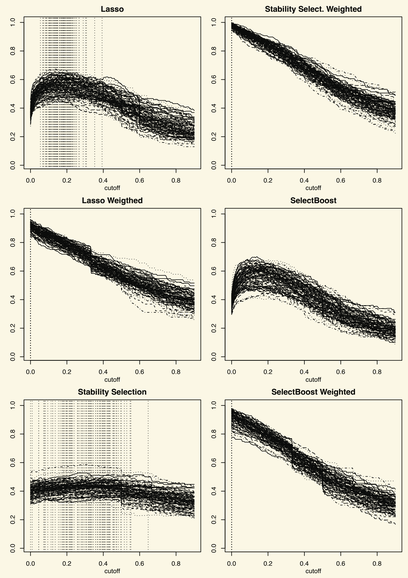

Fig. 1. Evolution of the F-score as a function of the cutoff used as a threshold to set as non-zero any inferred coefficient in network reverse engineering algorithm with or without relevant biological weights.

References

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